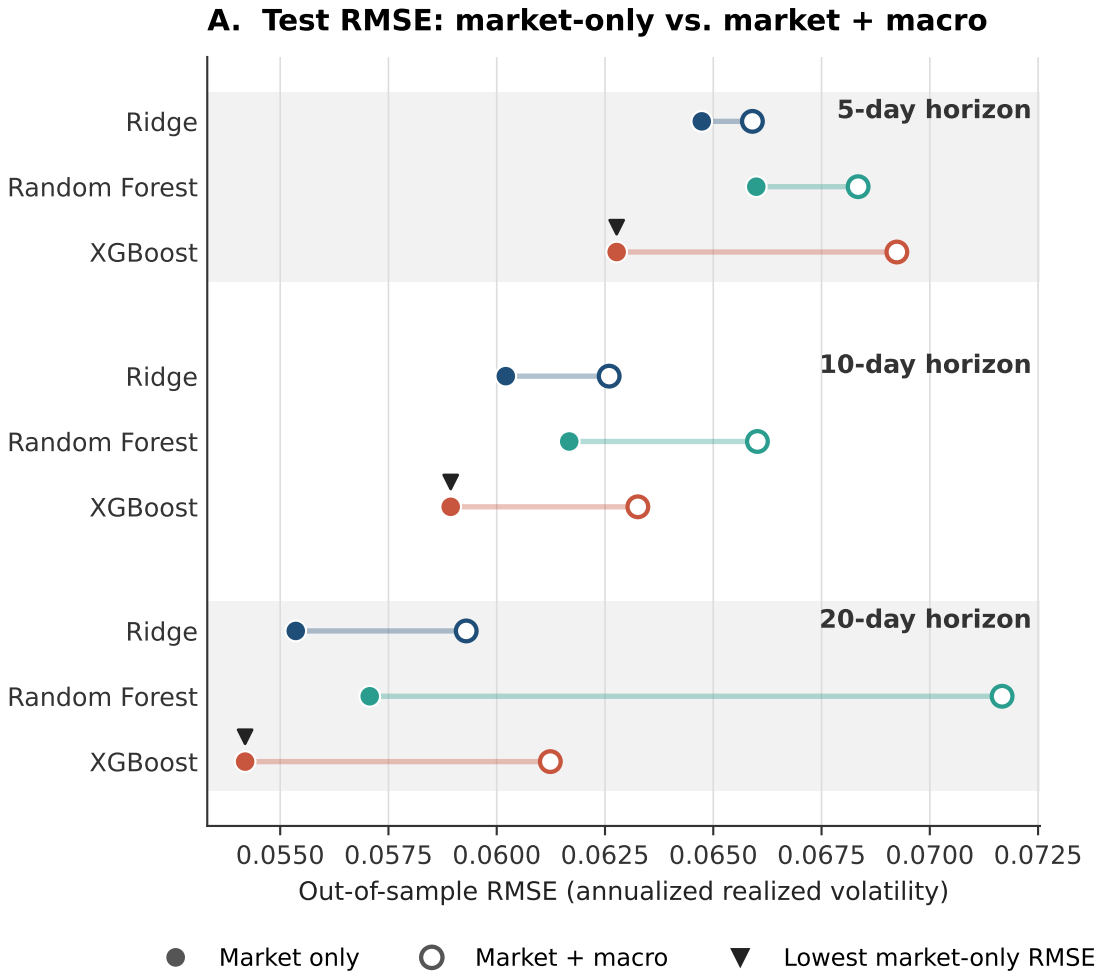
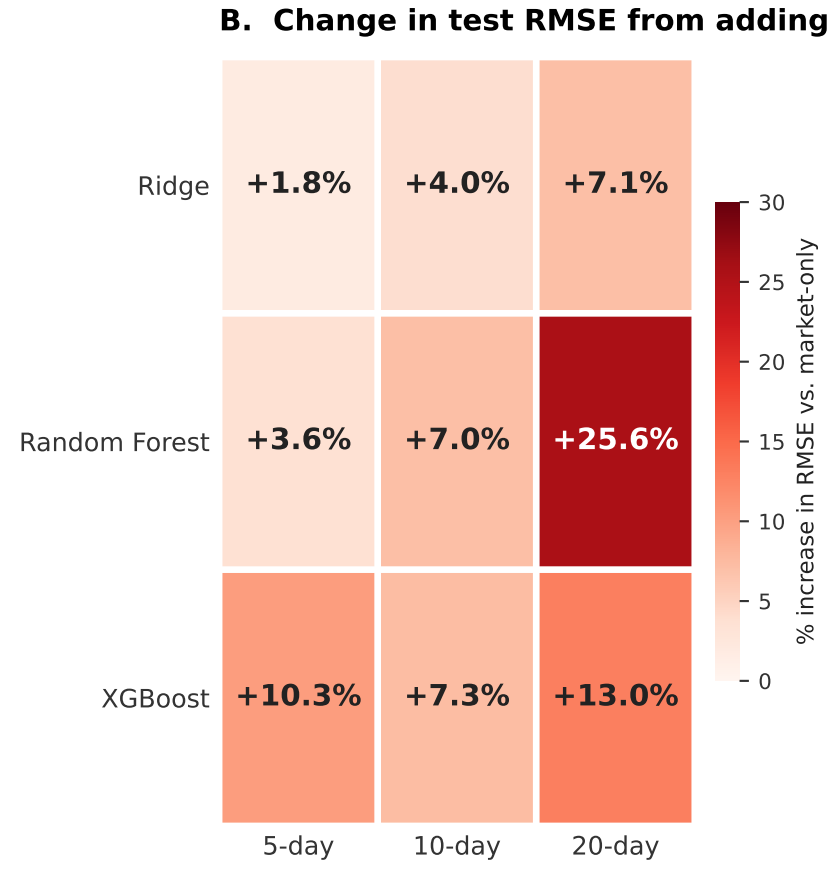
Macro features do not improve S&P 500 realized-volatility forecasts

Training through Dec 2018, validation 2019–2021, test Jan 2022 onward. Adding macro raises test RMSE for every model and horizon.





Note: RMSE on forward realized volatility over 5, 10 and 20 trading days. Macro set adds Treasury yields, 10Y–2Y spread, WTI oil, HY credit spread and fed funds.